

$$\Omega = (r/R)\omega$$

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Ans: D

By conservation of energy,

$$KE_i + GPE_i = KE_f + GPE_f$$

$$0 + \left(-\frac{GMm}{6R} \right) = \frac{1}{2}mv^2 + \left(-\frac{GMm}{R} \right)$$

$$v = \sqrt{\frac{5GM}{3R}}$$

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Ans: D